Lecture notes on advanced linear algebra

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Contents

Chapter 0

Review from linear algebra

0.1 Linear equations and systems

Definition 0.1.1. A **system of linear equations** is a collection of linear equations in the same variables $x_1, \dots, x_n$, that is

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

A system of linear equations is called **consistent** if it has either one of infinitely many solutions. If a system has no solutions it is called **inconsistent**.

When a system has more equations than variables we call it **overdetermined**. If there are more variables than equations we call it **underdetermined**.

Definition 0.1.2. The **coefficient matrix**, A, is a rectangular array having m rows and n columns where the coefficients of the system are recorded. The right–hand side vector, b, is a column that records the right–hand sides of the equations. That is,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}.$$

For any $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$, we refer to a_{ij} as the (i, j)th entry of the matrix A and to b_j as the jth entry of the vector b.

We can combine the above information into one matrix, called the augmented matrix of the linear system, as follows:

$$[A \mid b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array}\right].$$

Definition 0.1.3 (Operations that preserve solution set)**.** The following three operations preserve the solution set of a system of linear equations

1. Add a multiple of one equation to another equation.
2. Multiply any equation by a nonzero number.
3. Permute the equations as needed.

Definition 0.1.4. A matrix is said to be in echelon form (EF) if the following conditions are satisfied:

1. Any zero rows (i.e., all row entries being equal to 0) are at the bottom.
2. Every nonzero row has a leading (leftmost) nonzero entry, called the pivot, which is in a column to the right of the pivot in any row above it. We refer to a column that contains a pivot as a pivot column and the location of the pivot as the pivot position in that column.
3. All the column entries below a pivot are equal to 0.

We say the matrix is in reduced echelon form (REF) if it is in EF and in addition

1. All pivots are equal to 1.
2. All the column entries above a pivot are equal 0.

With the above set of definitions we can systematically solve a system of equations. This is generally taught using Gaussian Elimination. This algorithm is effective and easy to follow, however it is computationally inefficient and generally not used by computers. Matlab for example uses a form of LU decomposition which we will see later this semester. Solving a system of equations can be done with a myriad of algorithms in $O(n^3)$ time, this means that even massive matrices (millions by millions) can generally be solved in a short period of time.

0.2 Matrix notation and basic properties

A matrix is denoted using square brackets and with a capital letter. If a matrix is said to be m by n, then we mean m rows by n columns.

The notation $A = [a_{ij}]$ for $i = 1, \ldots, m$ and $j = 1, \ldots, n$. Means A is an m by n matrix whose ith jth entry is a_{ij}. This is useful when we want a generic matrix where we can reference specific entries. Another common notation is

$$A = [a_1 \mid a_2 \mid \cdots \mid a_n]$$

where a_i is a column vector of A.

A matrix is called square if and only if $m = n$. Two matrices are equal if their dimensions (m, n) are equal and all their entries are equal. The diagonal entries of a matrix are when the row and columns are equal. The off-diagonal entries are the rest.

Denote $\mathsf{M}_{n\times m}(S)$ to be the set of $n \times m$ matrices whose entries are pulled from the set S. If $n = m$, then we will write $\mathsf{M}_n(S)$. Another frequent notation for a set of matrices is $S^{n\times n}$, this is most common with $\mathbb{R}^{m\times n}$.

A matrix is called **upper triangular** if all the entries below the main diagonal are zero. A matrix is called diagonal if all off-diagonal entries are zero.

Definition 0.2.1 (Matrix addition and scalar multiplication)**.** Let $A = [a_{ij}]$ and$B = [b_{ij}]$ be two $m \times n$ matrices, and let r be a scalar.

The matrix sum of A and B is an $m\times n$ matrix $C = A+B = [c_{ij}]$ such that

$$c_{ij} = a_{ij} + b_{ij} \text{ for all } i = 1, 2, \ldots, m \text{ and all} j = 1, 2, \ldots, n.$$

The scalar product of A by r is an $m \times n$ matrix $C = rA = [c_{ij}]$ such that

$$c_{ij} = ra_{ij} \text{ for all } i = 1, 2, \ldots, m \text{ and all } j = 1, 2, \ldots, n.$$

Theorem 0.2.2 (Basic properties of matrix addition)**.** *Let A, B, C be matrices and r be a scalar.*

1. $A + B = B + A$ *(commutative property).*
2. $(A + B) + C = A + (B + C)$ *(associative property).*
3. $A + 0 = A = 0 + A$ *(we say 0 is the additive identity or the neutral of addition).*
4. $A + (-A) = 0 = (-A) + A$ *(we say $-A$ is the additive inverse of A).*
5. $(rs)A = (sr)A = s(rA) = r(sA)$.
6. $r(A + B) = rA + rB$ *and* $(r + s)A = rA + sA$ *(distributive property).*
7. $1A = A$.
8. *If* $A + B = C$*, then* $A = C - B$ *and* $B = C - A$.
9. *If* $A + C = B + C$*, then* $A = B$ *(cancellation law).*

With the above properties we shall see that matrices form a vector space under addition and scalar multiplication.

Definition 0.2.3 (Matrix vector multiplication)**.** Given an $m \times n$ matrix A whose columns are $a_1, a_2, \ldots, a_n$, and given a column vector x whose entries are $x_1, x_2, \ldots, x_n$, the product Ax is defined as the linear combination

$$Ax = [a_1 \mid a_2 \mid \cdots \mid a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n.$$

Definition 0.2.4 (Matrix Multiplication)**.** Let A be an $m \times k$ matrix and B a $k \times n$ matrix whose columns are $b_1, b_2, \ldots, b_n$. Then, the product of matrices AB is defined as the $m \times n$ matrix whose columns are Ab_1, Ab_2, $\ldots$, Ab_n, that is,

$$AB = A[b_1 \mid b_2 \mid \cdots \mid b_n] = [Ab_1 \mid Ab_2 \mid \cdots \mid Ab_n].$$

It is important to think about the above theoretical importance of what matrix multiplication is, each column of AB is a linear combination of the columns of A using the entries of the corresponding column of B. So one way to think about this is that the matrix B is the weights of how we want the columns of A to be mixed and combined.

A more practical way of actually computing the entries of the products is for $A = [a_{ij}]$ and $B = [b_{ij}]$, then the ith jth entry of AB is

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj} = \sum_{h=1}^{k} a_{ih}b_{hj}.$$

Definition 0.2.5 (Commutative matrices)**.** We say that two matrices A and B commute if AB and BA are both well-defined products and $AB = BA$.

Definition 0.2.6 (Identity and zero matrix)**.** We call the matrix I which is a diagonal matrix of all 1's the identity matrix. That is $I = \mathrm{diag}(1, 1, \ldots, 1)$.

We denote the zero matrix 0 or 0_n to be the matrix of all zero entries.

Definition 0.2.7 (Trace)**.** The trace of a matrix is the sum of the diagonal entries. Symbolically if $A = [a_{ij}]$ is a matrix, then

$$\mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Definition 0.2.8 (Transpose)**.** The transpose of a matrix swaps its rows and columns. Symbolically if $A = [a_{ij}]$, then $A^\top = [a_{ji}]$.

Theorem 0.2.9 (Matrix multiplication properties)**.** *Let A, B, C be matrices, then*

1. $AI = IA = A$.
2. $A0 = 0A = 0$.
3. $A(B + C) = AB + AC$ *and* $(B + C)A = BA + CA$.

4. *$A(BC) = (AB)C$.*

5. *$tr(AB) = tr(BA)$.*

6. *$(AB)^\top = B^\top A^\top$.*

7. *$AB = 0$ does **not** imply that either A or B is the zero matrix.*

Note that the last property of matrix multiplication means that generally division of matrices is not well defined.

Definition 0.2.10. Let A be a square matrix, then $A^k = AAA\dots A$, so powers are the standard repeated multiplication. By convention $A^0 = I$. We will talk later about negative powers.

Definition 0.2.11. A matrix A is called **symmetric** if $A = A^\top$. The matrix A is instead called **skew-symmetric** if $A^\top = -A$.

0.2.1 Exercises

1. For matrices
$$A = \begin{bmatrix} 1 & 1 & 5 \\ 7 & 3 & -2 \\ 15 & 1 & -3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 2 & 0 \\ 4 & 0 & 3 \\ -7 & 1 & 2 \end{bmatrix}.$$
Find the trace of the following
$$A, A^\top, B, \text{ and } AB.$$

2. For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, prove that $A^2 - (a+d)A + (ad-bc)I = 0$.

3. Let $A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$. Find all matrices $X = \begin{bmatrix} 1 & x \\ y & z \end{bmatrix}$ such that $AX = XA$.

4. Find three 2×2 matrices $A(\neq 0)$, B, and C such that $AB = AC$ and $B \neq C$.

5. If A, B are both n by n matrices with real entries such that $\text{tr}(A) = 3$ and $\text{tr}(B) = 5$, then what can we say, if anything, about $\text{tr}(AB)$?

6. Find two matrices A, B such that $AB = 0$, but $A \neq 0$ and $B \neq 0$.

7. Let A and B be two symmetric matrices such that $A^2 + B^2 = 0$. Prove that $A = B = 0$. Hint: Use the trace of A^2.

8. Let A and B be two $n \times n$ matrices such that $A^2 = A$, $B^2 = B$, and $(A+B)^2 = A + B$. Prove that $AB = BA = 0$.

0.2.2 Solutions

1. For matrices
$$A = \begin{bmatrix} 1 & 1 & 5 \\ 7 & 3 & -2 \\ 15 & 1 & -3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 2 & 0 \\ 4 & 0 & 3 \\ -7 & 1 & 2 \end{bmatrix}.$$
Find the trace of the following
$$A, A^\top, B, \text{ and } AB.$$
The trace is the sum of the diagonal entries, so the trace of A is $1+3+-3 = 1$. The trace of $A^\top$ is still 1 since taking the transpose does not affect the main diagonal. The trace of B is $3+0+2=5$. Finally to find the trace of AB we can solve for the three diagonal entries which are $1(3)+1(4)+5(-7) = -23$, $7(2)+3(0)+-2(1) = 12$, and $15(0)+1(3)+-3(2) = -3$. So the trace is -14.

2. For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, prove that $A^2 - (a+d)A + (ad-bc)I = 0$.

To prove this we can take each piece of the sum on the left individually, giving
$$A^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}.$$
$$(a+d)A = \begin{bmatrix} a^2+ad & ab+bd \\ ac+cd & ad+d^2 \end{bmatrix}$$
$$(ad-bc)I = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}.$$
Now taking the sum we can see that each entry gives 0.

3. Let $A = \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$. Find all matrices $X = \begin{bmatrix} 1 & x \\ y & z \end{bmatrix}$ such that $AX = XA$.

To start we can find AX and XA which are
$$AX = \begin{bmatrix} 4+2y & 4x+2z \\ 3+y & 3x+z \end{bmatrix},$$
$$XA = \begin{bmatrix} 4+3x & 2+x \\ 4y+3z & 2y+z \end{bmatrix}.$$
Now we can subtract these two to get the following 4 equations
$$\begin{aligned} -3x+2y &= 0 \\ 3x+2z-2 &= 0 \\ -3y-3z+3 &= 0 \\ 3x-2y &= 0. \end{aligned}$$

Notice that the last equation is just the negative of the first so we can ignore it. For the rest we can solve this with a standard coefficient matrix giving

$$\begin{bmatrix} -3 & 2 & 0 & 0 \\ 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 \end{bmatrix}.$$

Solving this gives the constraint that $3x+2z = 2$. Thus, there are infinitely many solutions.

4. Find three 2×2 matrices $A(\neq 0)$, B, and C such that $AB = AC$ and $B \neq C$.

   There are a lot of ways to approach this problem. One option is to just pick a random nice matrix A, then solve for B and C. For this we can pick

   $$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \ B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}, \text{ and } \ C = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}.$$

   Taking both sides gives

   $$AB = \begin{bmatrix} b_1 + b_3 & b_2 + b_4 \\ b_1 + b_3 & b_2 + b_4 \end{bmatrix} = \begin{bmatrix} c_1 + c_3 & c_2 + c_4 \\ c_1 + c_3 & c_2 + c_4 \end{bmatrix} = AC.$$

   This gives that $b_1 + b_3 = c_1 + c_3$ and $b_2 + b_4 = c_2 + c_4$. Now we can just pick some values such that B and C are not equal. In this case let $b_1 = b_2 = 1$, $b_3 = b_4 = -1$, $c_1 = c_2 = 2$, and $c_3 = c_4 = -2$. Then we have satisfied the sums, we clearly have that $B \neq C$ and $AB = AC = 0$.

5. If A, B are both n by n matrices with real entries such that $\mathrm{tr}(A) = 3$ and $\mathrm{tr}(B) = 5$, then what can we say, if anything, about $\mathrm{tr}(AB)$?

   We can't say anything about $\mathrm{tr}(AB)$. Consider

   $$A = \begin{bmatrix} x & & \\ & -x+3 & \\ & & 0 \end{bmatrix} \text{ and } \ B = \begin{bmatrix} y & & \\ & 0 & \\ & & -y+5 \end{bmatrix},$$

   then we nicely have the trace requirements for A and B. However $\mathrm{tr}(AB) = xy$ and since x and y are arbitrary numbers we can make this trace whatever we want. Thus there is nothing nice we can say about the trace of them together.

6. Find two matrices A, B such that $AB = 0$, but $A \neq 0$ and $B \neq 0$.

   We actually did this as part of the solution for problem 4. Take

   $$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \text{ and } \ B = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

   Then we have $AB = 0$.

7. Let A and B be two symmetric matrices such that $A^2 + B^2 = 0$. Prove that $A = B = 0$.

Proof. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ for $i, j \in \{1, \ldots, n\}$. Now if $A^2+B^2 = 0$ then we have $\operatorname{tr}(A^2 + B^2) = 0$ which gives that $\operatorname{tr}(A^2) + \operatorname{tr}(B^2) = 0$. Lets look at each trace individually

$$\operatorname{tr}(A^2) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}a_{ji}.$$

Since A is symmetric we have $a_{ij} = a_{ji}$ which simplifies the equation to

$$\operatorname{tr}(A^2) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2.$$

A similar calculation with B gives the following

$$\operatorname{tr}(A^2) + \operatorname{tr}(B^2) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2 + b_{ij}^2 = 0.$$

Looking at what we have here all of the entries in the above sum are being squared and are thus nonnegative. If we sum a bunch of nonnegative terms together and get zero we must have that all of the terms are zero. Therefore $A = 0$ and $B = 0$. $\square$

8. Let A and B be two $n \times n$ matrices such that $A^2 = A$, $B^2 = B$, and $(A + B)^2 = A + B$. Prove that $AB = BA = 0$.

Proof. Let $A, B \in \mathsf{M}_n$ such that $A^2 = A$, $B^2 = 0$, and $(A+B)^2 = A+B$, then expand the binomial normally giving

$$(A + B)^2 = A^2 + AB + BA + B^2 = A + AB + BA + B = A + B.$$

Thus we have that $AB + BA = 0$ or that $AB = -BA$. Now multiply this by A on the left giving $A^2B = -ABA$ which simplifies to $AB = -ABA$. Additionally multiply this by A on the right giving $ABA = -BA^2$ which becomes $BA = -ABA$. Thus $BA = -ABA$ and $AB = -ABA$ so $AB = BA$. However we have $AB = -BA$ and $AB = BA$ thus $AB = 0$ and $BA = 0$. $\square$

Chapter 1

Vector spaces

1.1 Definition

Definition 1.1.1 (Field)**.** A *field* is a set with two binary operations, addition (denoted by $a+b$) and multiplication (denoted by ab), such that for all $a, b, c \in F$:

1. $a + b = b + a$.
2. $(a + b) + c = a + (b + c)$.
3. There is an additive identity 0; that is $0 + a = a$ for all $a \in R$.
4. There is an element $-a$ in R such that $a + (-a) = 0$.
5. $a(bc) = (ab)c$.
6. $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.
7. $ab = ba$
8. $\exists 1 \in F$ such that $1a = a1 = a$.
9. If $a \neq 0$, then there exists $a^{-1} \in F$ such that $aa^{-1} = 1$.

Example 1.1.2. The rational ($\mathbb{Q}$), real ($\mathbb{R}$), and complex numbers ($\mathbb{C}$) form fields.

The integers ($\mathbb{Z}$) however, do not since they have no multiplicative inverses. $\triangle$

Example 1.1.3. Let

$$\begin{aligned} \mathbb{Z}_3[i] &= \{a + bi \mid a, b \in \mathbb{Z}_3\} \\ &= \{0, i, 2i, 1, 1 + i, 1 + 2i, 2, 2 + i, 2 + 2i\}. \end{aligned}$$

Where the addition and multiplication operations are as they are for complex numbers mod 3. In particular note that $-1 = 2$. With these operations $\mathbb{Z}_3[i]$ forms a field with 9 elements. $\triangle$

Definition 1.1.4 (Vector space)**.** A set V is said to be a vector space over a field F if V has two operations addition and scalar multiplication such that for all $u, v \in V$ and $a \in F$ they satisfy the following

1. $u + v = v + u$
2. $(u + v) + w = u + (v + w)$
3. There exists a $0 \in V$ such that $0 + u = u + 0 = u$.
4. There exists a $-u$ such that $u + (-u) = 0$.
5. $a(v + u) = av + au$
6. $(a + b)v = av + bv$
7. $a(bv) = (ab)v$
8. $1v = v$.

The elements in V are called vectors while the elements from F are called scalars.

If the field for a vector space is $\mathbb{R}$ we call it a real vector space and if the field is $\mathbb{C}$ we call it a complex vector space. These will be by far our most common vector spaces.

Example 1.1.5. The set

$$\mathbb{R}^n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in \mathbb{R}\}$$

is a vector space over $\mathbb{R}$. Where the operations are elementwise addition and scalar multiplication. △

Example 1.1.6. The set $\mathsf{M}_2(\mathbb{Q})$ of 2 by 2 matrices with entries from $\mathbb{Q}$ is a vector space over $\mathbb{Q}$. The operations are

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} + \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 & a_2 + b_2 \\ a_3 + b_3 & a_4 + b_4 \end{bmatrix}$$

and

$$b \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} = \begin{bmatrix} ba_1 & ba_2 \\ ba_3 & ba_4 \end{bmatrix}.$$

△

Example 1.1.7. The set $\mathbb{Z}_p[x]$ of polynomials with coefficients from $\mathbb{Z}_p$ is a vector space over $\mathbb{Z}_p$, where p is a prime.

More generally, if F is a field and $F[x]$ are the polynomials with coefficients from F, then $F[x]$ is a vector space. △

Theorem 1.1.8. *Let V be a vector space.*

1. *The zero vector of V is unique.*
2. *The negative of a vector $v \in V$ is unique.*
3. $-v = (-1)v$ *for all* $v \in V$.
4. *If* $rv = 0$ *for some* $v \in V$ *and* $r \in F$, *then either* $v = 0$ *or* $r = 0$.

Proof. Assume there is another zero element $u \in V$, then by the properties of being a vector space $u + 0 = 0$ and $u + 0 = u$. Thus $u = 0$ giving that there was only one zero element.

Let $v \in V$ and assume that u_1 and u_2 in V such that $v + u_1 = 0$ and $v + u_2 = 0$, then

$$u_1 = u_1 + 0 = u_1 + (v + u_2) = (u_1 + v) + u_2 = 0 + u_2 = u_2.$$

For any $v \in V$ we have

$$[1 + (-1)]v = 0 \implies 1v + (-1)v = 0 \implies (-1)v = -v.$$

Suppose $rv = 0$ for some $r \in F$ and $v \in V$, then proceed by cases. If $r = 0$, then we are done. If $r \neq 0$, then

$$v = 1v = \frac{1}{r}rv = \frac{1}{r}0 = 0.$$

□

1.1.1 Exercises

1. Prove that $V = \{(x, y) \in \mathbb{R}^2 \mid y = 5x^2\}$ is not a vector space.
2. Prove that $V = \{(x, y, z) \in \mathbb{C}^3 \mid z = 2x + 3y + 1\}$ is not a vector space.
3. Prove that the set of all nonnegative infinite sequences $V = \{(x_1, x_2, \dots) \mid x_i \geq 0\}$ is not a vector space.

1.1.2 Solutions

1. Prove that $V = \{(x, y) \in \mathbb{R}^2 \mid y = 5x^2\}$ is not a vector space.

   Consider the vector $a = (1, 5) \in V$, then $a + a = (2, 10)$ is not in V. Thus V is not a vector space since it is not closed under addition.

2. Prove that $V = \{(x, y, z) \in \mathbb{C}^3 \mid z = 2x + 3y + 1\}$ is not a vector space.

   This set V is missing the zero vector.

3. Prove that the set of all nonnegative infinite sequences $V = \{(x_1, x_2, \dots) \mid x_i \geq 0\}$ is not a vector space.

   Let $a = (1, 1, \dots) \in V$, then the additive inverse is $-a = (-1)a = (-1, -1, \dots)$ which is not in V.

1.2 Subspace

Definition 1.2.1 (Subspace)**.** Let V be a vector space over a field F and let U be a nonempty subset of V. We say that U is a *subspace* of V if U is also a vector space over F under the operations of V.

Theorem 1.2.2. *Let U be a nonempty subset of a vector space V. Then, U is a subspace of V if and only if*

$$ru + sw \in U$$

for every $u, w \in U$ and $r, s \in \mathbb{R}$.

Proof. Let U be a subset of a vector space V and suppose that $ru + sw \in U$ for every $u, w \in U$ and $r, s \in \mathbb{R}$. For $r = s = 1$ we get that U is closed under vector addition. For $s = 0$ we get that U is closed under scalar multiplication. Hence U is a subspace of V.

Conversely, suppose that $U \leq V$, then U is closed under scalar multiplication and vector addition. Thus for any $r, s \in \mathbb{R}$ and $u, w \in U$ we have that ru and sw are in U which means their sum $ru + sw \in U$. $\square$

Definition 1.2.3. Let V be a vector space and let U and W be two subspaces of V. The **intersection of subspaces** U and W is the set

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}$$

and the **sum of subspaces** U and W is the set

$$U + W = \{v \in V : v = u + w, \ u \in U, \text{ and } w \in W\}.$$

If $U \cap W = \{0\}$, then the sum $U + W$ is called the **direct sum of subspaces** and is denoted $U \oplus W$.

In the case where $V = U \oplus W$, W is known as the **complementary subspace** of U.

Theorem 1.2.4. *Let V be a vector space, and let U and W be two subspaces of V. The intersection $U \cap W$ and the sum $U + W$ are also subspaces of V.*

Proof. To do as exercise. $\square$

Theorem 1.2.5. *Let V be a vector space, and let U and W be two subspaces of V. Then $V = U \oplus W$ if and only if every $v \in V$ can be uniquely written as a sum $v = u + w$ with $u \in U$ and $w \in W$.*

Proof. Suppose that $V = U \oplus W$. If there exists $u', u \in U$ and $w', w \in W$ such that $v = u' + w'$ and $v = u + w$, then

$$\begin{aligned} v = u + w = u' + w' &\implies u - u' = w' - w \in U \cap W = \{0\} \\ &\implies u' = u \quad \text{and} \quad w' = w. \end{aligned}$$

Hence the decomposition $v = u + w$ is unique.

For the converse, suppose that any vector $v \in V$ can be uniquely written as a sum $v = u + w$ with $u \in U$ and $w \in W$. It is obvious that $V = U + W$ and therefore we must only prove that $U \cap W = \{0\}$. Assume for contradiction that this is not true, that is $U \cap W$ contains some vector x. Then by the uniqueness of the decomposition of x we can write $x = x + 0$ where $x \in U$ and $x = 0 + x$ where $x \in V$, but this implies $x = 0$. Thus we have a contradiction, giving that $U \cap W = \{0\}$. $\square$

Example 1.2.6. Consider the vector space $\mathbb{R}^3$ and its subsets

$$\begin{aligned} V &= \{(x, y, z) \in \mathbb{R}^3 : z = 0\} \\ U &= \{(x, y, z) \in \mathbb{R}^3 : x = 0\} \\ W &= \{(x, y, z) \in \mathbb{R}^3 : x = y = 0\}. \end{aligned}$$

We can see that each of these subsets are in fact subspaces of $\mathbb{R}^3$ since they are closed under the two operations.

Looking at the intersections we get

$$\begin{aligned} V \cap U &= \{(x, y, z) \in \mathbb{R}^3 : x = z = 0\} \\ V \cap W &= \{(x, y, z) \in \mathbb{R}^3 : x = y = z = 0\} = \{0\} \\ U \cap W &= \{(x, y, z) \in \mathbb{R}^3 : x = y = 0\} = W. \end{aligned}$$

Following the above and that $V + W = \mathbb{R}^3$ we see that $V \oplus W = \mathbb{R}^3$. Furthermore, as you will prove since $U \cap W = W$ we know that U is a subspace of W. $\triangle$

1.2.1 Exercises

1. Explain why the following subsets of $\mathbb{R}^2$ are not subspaces of $\mathbb{R}^2$:
   (a) $V_1 = \{(x, y) \in \mathbb{R}^2 : x \geq y\}$,
   (b) $V_2 = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}$,
   (c) $V_3 = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$.
2. Prove Theorem 1.2.4.
3. Prove that the sets

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid x = 2y\},$$
$$U = \{(x, y, z) \in \mathbb{R}^3 \mid x = -3z\}$$

   are subspaces of $\mathbb{R}^3$. Then, describe their intersection $V \cap U$.

1.2.2 Solutions

1. Explain why the following subsets of $\mathbb{R}^2$ are not subspaces of $\mathbb{R}^2$:
   (a) $V_1 = \{(x, y) \in \mathbb{R}^2 : x \geq y\}$,
   Let $v_1 = (2, 1)$ and $v_2 = (2, -1)$ which are both in V_1. Now $v_1 - v_2 = (0, 2)$ which is not in V_1 so it is not closed under vector addition.
   Another way to solve this is just to take v_2 and consider its additive inverse.
   (b) $V_2 = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}$,
   Let $v_1 = v_2 = (1, 1)$ which are both in V_2, then $v_1 + v_2 = (2, 2) \notin V_2$. Thus V_2 is not closed under addition.
   (c) $V_3 = \{(x, y) \in \mathbb{R}^2 : x + y = 1\}$.
   Let $v = (2, -1) \in V_3$, then $0v = (0, 0) \in V_3$. Thus V_3 is not closed under scalar multiplication.
   You can also argue that V_3 is missing the zero vector.
2. Prove Theorem 1.2.4.

Proof. Given a vector space V with field F and with subspaces U and W we want to show that the intersection and sum of U and W are subspaces of V. First consider the intersection, let $u, v \in U \cap W$ and $r, s \in F$. Now $ru + sv \in U$ since $u, v \in U$ and U is a subspace. By the same logic $ru + sv \in W$. Thus $ru + sv \in U \cap W$ giving that $U \cap W$ is a subspace.

For the sum let $v, u \in U + W$, then we can write $v = v_1 + v_2$ and $u = u_1 + u_2$ where $v_1, u_1 \in U$ and $v_2, u_2 \in W$. Now we have

$$rv + su = rv_1 + rv_2 + su_1 + su_2.$$

Since U and W are both subspaces we have $rv_1 + su_1 \in U$ and $rv_2 + su_2 \in W$ giving that $rv_1 + rv_2 + su_1 + su_2 \in U + W$. $\square$

3. Prove that the sets
$$V = \{(x, y, z) \in \mathbb{R}^3 \mid x = 2y\},$$
$$U = \{(x, y, z) \in \mathbb{R}^3 \mid x = -3z\}$$
are subspaces of $\mathbb{R}^3$. Then, describe their intersection $V \cap U$.

Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$ and $r, s \in \mathbb{R}$, then we want to show that $r(x_1, y_1, z_1) + s(x_2, y_2, z_2) \in V$. To do this it must satisfy the restriction, which we can see from
$$rx_1 + sx_2 = r(2y_1) + s(2y_2) = 2(ry_1 + sy_2).$$
Thus V is a subspace of $\mathbb{R}^3$.

Similar for U let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in U$ and $r, s \in \mathbb{R}$, then from the linear combination we get the condition
$$rx_1 + sx_2 = r(-3z_1) + s(-3z_2) = -3(rz_1 + sz_2).$$
When we take the intersection of subspaces defined by linear constraints we get both of their constraints. So in this case
$$V \cap U = \{(x, y, z) \in \mathbb{R}^3 \mid x = 2y \text{ and } x = -3z\}.$$
Note also that because we don't have that $V \subseteq U$ or $U \subseteq V$ the intersection is a proper subset.

1.3 Linear combinations

Definition 1.3.1. Let V be a vector space and let $v_1, v_2, \ldots, v_k \in V$. For any k scalars $r_1, r_2, \ldots, r_k \in \mathbb{R}$, the vector
$$v = r_1v_1 + r_2v_2 + \cdots + r_kv_k \in V$$
is called a **linear combination** of $v_1, v_2, \ldots, v_k$ with **coefficients** $r_1, r_2, \ldots, r_k$. The set of all linear combinations of $v_1, v_2, \ldots v_k$ is denoted by
$$\text{span}\{v_1, v_2, \ldots, v_k\} = \{v = r_1v_1 + r_2v_2 + \cdots + r_kv_k \in V : r_1, r_2, \ldots, r_k \in \mathbb{R}\},$$
and it is called the **span** (or linear hull) of $v_1, \ldots, v_k$. The vectors $v_1, \ldots, v_k$ are called the **generators** of this span.

Theorem 1.3.2. *Let V be a vector space, and let $v_1, \ldots, v_k \in V$. Then span$\{v_1, \ldots, v_k\}$ is a subspace of V.*

Not only is the span of vectors a subspace of the original vector space, but it represents the smallest subspace that contains those vectors.

Theorem 1.3.3. *Let U and W be two subspaces of a vector space V, then their union*
$$U \cup W = \{v \in V : v \in U \text{ or } v \in W\}$$
is a subspace of V if and only if $U \leq W$ or $W \leq U$.

Proof. To do on homework. ☐

1.4 Linear Independence

Definition 1.4.1. Let V be a vector space, and let $v_1, v_2, \ldots, v_k \in V$. The set $\{v_1, \ldots, v_k\}$ is called **linearly dependent** if at least one of its vectors can be written as a linear combination of the others. Otherwise the set is called **linearly independent**.

Theorem 1.4.2. *Let V be a vector space over a field F, and let $v_1, \ldots, v_k \in V$. The set $\{v_1, \ldots, v_k\}$ is linearly independent if and only if it satisfies the following condition*

$$r_1 v_1 + \cdots + r_k v_k = 0 \quad \textit{for} \quad r_1, \ldots, r_k \in F \implies r_1 = \cdots = r_k = 0.$$

Example 1.4.3. Lets prove that the vectors $(1,1,1), (1,1,-1), (1,-1,-1) \in \mathbb{R}^3$ are linearly independent. First we have that

$$r_1(1,1,1) + r_2(1,1,-1) + r_3(1,-1,-1) = 0.$$

This gives the equations

$$\begin{aligned} r_1 + r_2 + r_3 &= 0 \\ r_1 + r_2 - r_3 &= 0 \\ r_1 - r_2 - r_3 &= 0. \end{aligned}$$

From the first equation $r_3 = -r_1 - r_2$, substituting this into the other two gives

$$\begin{aligned} 2r_1 + 2r_2 &= 0 \\ 2r_1 &= 0. \end{aligned}$$

From the second equation we see that $r_1 = 0$, but this implies from the first equation that $r_2 = 0$. Going back to the original we have that $r_3 = 0$. Thus the only linear combination that gives zero vector is with zero coefficients giving that these vectors are linearly independent. $\triangle$

Theorem 1.4.4. *Consider the vector space $\mathbb{R}^n$ and vectors $v_1, \ldots, v_k \in \mathbb{R}^n$. Form the $n \times k$ matrix A whose columns are the vectors $v_1, \ldots, v_k$ and compute an echelon form E of A. Then, $\{v_1, \ldots, v_k\}$ is linearly independent if and only if every column of E contains a pivot.*

1.4.1 Exercises

1. Prove that the vectors $v_1 = (1, -1)$ and $v_2 = (1, 2)$ span the vector space $\mathbb{R}^2$.

2. Prove that the vectors $v_1 = (1, 1, 1)$, $v_2 = (0, 2, 1)$, and $v_3 = (0, 0, 3)$ span the vector space $\mathbb{R}^3$.

3. Prove that $\operatorname{span}\{(1, 2, 3), (-1, 3, 2)\} = \operatorname{span}\{(1, 0, 1), (1, 1, 2)\}$.

4. Let V be a vector space, and let $u, v_1, v_2, \ldots, v_k \in V$. Prove that

$$\operatorname{span}\{u, v_1, \ldots, v_k\} = \operatorname{span}\{v_1, \ldots, v_k\}$$

   if and only if $u \in \operatorname{span}\{v_1, \ldots, v_k\}$.

5. Prove that if V is a vector space with vectors $v_1, \ldots, v_k$ and if $v_i = 0$ for some $i \in \{1, \ldots, k\}$, then the set $\{v_1, \ldots, v_k\}$ is linearly dependent.

6. Let V be a vector space and let $u, v, w \in V$. Prove that u, v, and w are linearly independent if and only if u, $u + v$, $u + v + w$ are linearly independent.

1.4.2 Solutions

1. Prove that the vectors $v_1 = (1, -1)$ and $v_2 = (1, 2)$ span the vector space $\mathbb{R}^2$.

   To show we span the vector space $\mathbb{R}^2$ we want to show that every vector in $\mathbb{R}^2$ can be achieved as a linear combination of v_1 and v_2. To do this let $(x, y) \in \mathbb{R}^2$, then for $r_1 v_1 + r_2 v_2 = (x, y)$ we need $r_1 + r_2 = x$ and $-r_1 + 2r_2 = y$. Solving gives that $r_1 = 2x - y$ and $r_2 = y - x$. Since these systems are consistent we know the vectors span.

2. Prove that the vectors $v_1 = (1, 1, 1)$, $v_2 = (0, 2, 1)$, and $v_3 = (0, 0, 3)$ span the vector space $\mathbb{R}^3$.

   For this problem we will take a different approach than the first one. We can instead make a matrix from the vectors which is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 3 \end{bmatrix}.$$

   Row reducing this gives the identity matrix which has a pivot in every row. Since the solution space stays the same through row reduction we know that these vectors span $\mathbb{R}^3$.

3. Prove that $\operatorname{span}\{(1, 2, 3), (-1, 3, 2)\} = \operatorname{span}\{(1, 0, 1), (1, 1, 2)\}$.

A direct way to show that these sets are equal is to show you can make the generators from one set using linear combinations from the other. That is we can take

$$\frac{3}{5}(1,2,3) - \frac{2}{5}(-1,3,2) = (1,0,1)$$
$$\frac{4}{5}(1,2,3) - \frac{1}{5}(-1,3,2) = (1,1,2).$$

Similarly for the other direction we get

$$-(1,0,1) + 2(1,1,2) = (1,2,3)$$
$$-4(1,0,1) + 3(1,1,2) = (1,1,2).$$

4. Let V be a vector space, and let $u, v_1, v_2, \ldots, v_k \in V$. Prove that

$$\operatorname{span}\{u, v_1, \ldots, v_k\} = \operatorname{span}\{v_1, \ldots, v_k\}$$

if and only if $u \in \operatorname{span}\{v_1, \ldots, v_k\}$.

If $\operatorname{span}\{u, v_1, \ldots, v_k\} = \operatorname{span}\{v_1, \ldots, v_k\}$, then we can write any vector in $\{u, v_1, \ldots, v_k\}$ as a linear combination of the vectors in $\{v_1, \ldots, v_k\}$. In particular, this means that

$$u = r_1 v_1 + r_2 v_2 + \cdots + r_k v_k$$

which implies that $u \in \operatorname{span}\{v_1, \ldots, v_k\}$.

Conversely if, $u \in \operatorname{span}\{v_1, \ldots, v_k\}$, then we can add u to the spanning set without changing it. Thus $\operatorname{span}\{u, v_1, \ldots, v_k\} = \operatorname{span}\{v_1, \ldots, v_k\}$

5. Prove that if V is a vector space with vectors $v_1, \ldots, v_k$ and if $v_i = 0$ for some $i \in \{1, \ldots, k\}$, then the set $\{v_1, \ldots, v_k\}$ is linearly dependent.

6. Let V be a vector space and let $u, v, w \in V$. Prove that u, v, and w are linearly independent if and only if u, $u + v$, $u + v + w$ are linearly independent.

1.5 Basis and dimension

Definition 1.5.1. Let $V \neq \{0\}$ be a vector space, and let $\{v_1, v_2, \ldots, v_n\}$ be a set of vectors in V. The set $\{v_1, \ldots, v_n\}$ is called a **basis** of the vector space V if

1. the set $\{v_1, \ldots, v_n\}$ is linearly independent, and
2. the vectors $v_1, \ldots, v_n$ span V.

Note that for infinite vector spaces, their are infinite fields since we can scale the basis to get a new one.

Theorem 1.5.2. *Let V be a vector space, and let $v_1, \ldots, v_n \in V$ be nonzero. The set $\{v_1, \ldots, v_n\}$ is a basis of V if and only if every vector $v \in V$ is written as a linear combination of $v_1, \ldots, v_n$ in a unique way.*

Theorem 1.5.3. *Let V be a vector space with a basis $\{v_1, v_2, \ldots, v_n\}$, and let $u_1, \ldots, u_m$ be linearly independent vectors of V with $m \leq n$. Then we can replace m vectors of the basis $\{v_1, \ldots, v_n\}$ by the vectors $u_1, \ldots, u_m$ such that the resulting set of n vectors is also a basis.*

Definition 1.5.4. Let $V \neq \{0\}$ be a vector space with a basis that has n elements, then the **dimension** of V is n. We denote this with

$$\dim(V) = n.$$

In this case V is referred to as a finite vector space.

If a basis of V requires an infinite number of vectors, then V is an infinite vector space.

By convention if $V = \{0\}$, then we say that V is zero dimensional.

Theorem 1.5.5. *Let V be a vector space with $\dim(V) = n$. The following hold:*

1. *If $m > n$, then the set $\{v_1, \ldots, v_m\}$ is linearly dependent.*
2. *If $\{v_1, \ldots, v_m\}$ is linearly independent, then $m \leq n$.*
3. *If $v_1, \ldots, v_m \in V$ span V, then $m \geq n$.*

Theorem 1.5.6. *Let V be a vector space with $\dim(V) = n$, and let U be a subspace of V. If $\dim(U) = n$, then $U = V$.*

Theorem 1.5.7. *Let V be a finite dimensional vector space, and let U and W be two subspaces of V. Then*

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

Note that by the above theorem if U and V are complementary, then we can drop the difference. That is $\dim(U \oplus W) = \dim(U) + \dim(W)$.

1.6 Linear transformations

Definition 1.6.1. Given two vector spaces V and W over fields F_V and F_W a **transformation** is a function

$$T : V \to W$$

that assigns to every $v \in V$ to a vector $u \in W$. We denote this with $T(v) = u$.

A transformation is called a **linear transformation** if it satisfies

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$ for every $v_1, v_2 \in V$
2. $T(cv) = cT(v)$ for every $v \in V$ and $c \in F_V$.

Definition 1.6.2. Let $T : V \to U$ be a be a transformation (function), then

1. The vector space V is called the **domain** and U is called the **codomain**.
2. The vector $u = T(v)$ is the **image** of v.
3. The set of all images is called the **range**, denoted $R(T)$.
4. The set of all vectors $v \in V$ such that $T(v) = 0$ is called the **nullspace**, denoted $\text{Null}(T)$.

1.6.1 Exercises

1. Are the following linear transformations
   (a) $T : \mathbb{R}^3 \to \mathbb{R}^3$ given by $T(u_1, u_2, u_3) = (u_1, u_2 + u_3, 0)$.
   (b) $T : \mathbb{R}^3 \to \mathbb{R}^3$ given by $T(u_1, u_2, u_3) = (1 + u_1, u_2, u_3)$.
   (c) $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by $T(u_1, u_2, u_3) = (2u_1, u_2 + u_3)$.
2. Find a basis and the dimension of the subspace of $\mathbb{R}^3$ that is spanned by the vectors $(1, -1, 0), (2, 1, 3), (1, 1, 2)$.
3. Find a basis and the dimension of the subspace of $\mathbb{R}^3$ that is spanned by the vectors $(1, 2, 1), (3, 3, 1), (1, 5, 3)$.
4. Consider the subspaces of $\mathbb{R}^3$

   $$U = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\} W = \{(x, y, z) \in \mathbb{R}^3 : 2x + 3y - z = 0\}$$

   Find a basis, and the dimension for U, W, $U + W$, and $U \cap W$.

1.7 Isomorphisms

Definition 1.7.1. Let V and W be vector spaces with $v_1, v_2 \in V$. Then a linear transformation $T : V \to W$ is called **one-to-one** if whenever $v_1 \neq v_2$ we have that $T(v_1) \neq T(v_2)$.

Definition 1.7.2. Let V and W be vector spaces. Then a linear transformation $T : V \to W$ is called **onto** if for all $w \in W$ there exists a $v \in V$ such that $T(v) = w$.

Theorem 1.7.3. *Let V and W be vector spaces. Then a linear transformation $T : V \to W$ is one to one if and only $Null(T) = \{0\}$.*

Definition 1.7.4. A linear transformation is called a **bijection** if it is one-to-one and onto.

Definition 1.7.5. Let V and W be vector spaces, then V and W are called **isomorphic** if there exists a bijective linear transformation T between them. This is denoted with $V \cong W$. This linear transformation T is called an **isomorphism** from V to W.

Theorem 1.7.6. *Let $T : V \to W$ be an isomorphism. Then $T^{-1} : W \to V$ is an isomorphism.*

Theorem 1.7.7. *Let V and W be vector spaces, let $T : V \to W$ be an isomorphism, and let $v_1, \dots, v_n$ be a basis for V. Then $T(v_1), \dots, T(v_n)$ is a basis for W.*

Theorem 1.7.8. *Suppose V and W are vector spaces over the same field. Then $V \cong W$ if and only if $\dim(V) = \dim(W)$.*

Furthermore if $\dim(V) = \dim(W)$ and $T : V \to W$ is a linear transformation, then the following are equivalent

1. *T is one-to-one,*
2. *T is onto, and*
3. *T is an isomorphism.*

This result allows us to always work over F^n for some field F as our vector space.

1.8 Standard matrix of a transformation

Chapter 2

Eigenvalues

2.1 Determinant

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2.3 Eigenvalues

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Chapter 3

Inner product spaces

Chapter 4

Matrix Factorizations

Chapter 5

Special matrix types